

cout << nums[0] << "\n");

Exp-2

- points array $points[i] = [x_i, y_i]$, return max. no. of points lying on same straight line.

Example = $[[1,1], [2,2], [3,3]]$

Output = 3

→ Algorithm:-

- Initialize result = 0
- For each point i :
 - Create a hashmap slopeCount
 - Initialize: overlap = 0 (duplicate points)
 - maxline = 0 (max points with same slope)
- For each point $j > i$:
 - Compute: $dx = x_j - x_i$
 $dy = y_j - y_i$
 - If both are zero:
 - Increment overlap
 - Continue
 - Normalize slope:
 - $g = \gcd(dx, dy)$
 - $dx /= g, dy /= g$
 - Store slope (dx, dy) in hashmap:
 - $slopeCount[(dx, dy)]++$
 - Update maxline
- Update result:
 $result = \max(result, maxline + overlap + 1)$
- Return result

→ Time Complexity: $O(n^2)$ Space Complexity: $O(n)$